Roll No. ....

TMA–302

B. TECH. (CE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

ENGINEERING MATHEMATICS–III

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Check the analyticity of the following functions : (CO1)

(i) $f(z) = |z|^2$ for all z

(ii) $f(z) = z^3$ for all z

OR

(b) Find the Fourier transform of the function : (CO2)

$$f(x) = \begin{cases} 1 - x^2, & |x| \le 1 \\ 0 \quad , & |x| \ge 1 \end{cases}$$

2. (a) Using Milne-Thomson method, find an analytic function $f(z) = u(x, y) + iv(x, y)$, where the real part of $f(z)$ is $u(x, y) = \sin x \cos y$. (CO1)

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OR

(b) Show that the function $f(z) = u + iv$, where :

$$f(z) = \begin{cases} \dfrac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

is not analytic at $z = 0$ although Cauchy-Riemann equations are satisfied at $z = 0$. (CO1)

3. (a) Write down the statement of Cauchy-Riemann's equation for Cartesian coordinate. Show that the function $f(z) = z + 2\bar{z}$ is not analytic anywhere in the complex plane. (CO1)

OR

(b) Find the Fourier transform of : (CO2)

$$f(x) = \begin{cases} 1 + \dfrac{x}{a}, & -a < x < 0 \\ 1 - \dfrac{x}{a}, & 0 < x < a \\ 0, & \text{otherwise} \end{cases}$$

4. (a) Evaluate the integral $\int_{(1,-1)}^{(2,1)} (2x + iy + 1)\, dz$ along the straight line from $z = 1 - i$ to $z = 2 + i$. (CO1)

OR

(b) Prove that $f(z) = 2x - x^3 + 3xy^2$ is harmonic function and find its harmonic conjugate. Also find the corresponding analytic function.

(CO1)

5. (a) Define analytic function. Show that the function $f(z) = z|z|$ is not analytic anywhere. (CO1)

OR

(b) Find whether the function $f(z) = u + iv$, where :

$$f(z) = \begin{cases} \dfrac{x^4(1+i) - y^4(1-i)}{x^3 + y^3}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

is analytic or not at $z = 0$. (CO1)